

Please see the Appendix at the end of this document first!

Common source stage

1 Resistive load

In the circuit of Figure 1, assume that $(W/L)_1 = 50/0.5$, $R_D = 2k\Omega$, and $\lambda = 0$.

- What is the small-signal gain if M_1 is in saturation and $I_D = 1$ mA?
- What input voltage places M_1 at the edge of the triode region? What is the small-signal gain under this condition?
- What input voltage drives M_1 into the triode region by 50 mV? What is the small-signal gain under this condition?

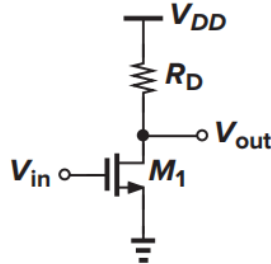


Figure 1

2 Diode-connected load

For the circuit of Figure 2, calculate the small-signal voltage gain if $(W/L)_1 = 50/0.5$, $(W/L)_2 = 10/0.5$, and $I_{D1} = I_{D2} = 0.5$ mA (body effect can be neglected). What is the gain if M_2 is implemented as a diode-connected PMOS device?

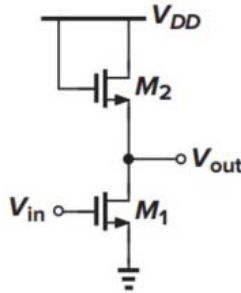


Figure 2

3 Current source load

For the circuit in Figure. 3:

- Draw the small signal model considering channel length modulation effect.
- Write the expression of small-signal gain and output resistance.
- Derive the expression for the maximum output swing.

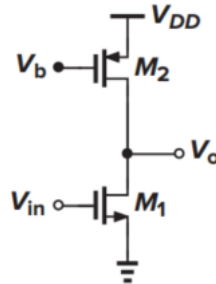


Figure 3

4 Class-AB stage

- Draw the small signal model for the circuit in Figure 4(a), and derive the expression of small-signal voltage gain.
- Derive the expression of small-signal power supply rejection ratio (PSRR) of the circuit in Figure 4(b). (Hint: $PSRR = A_v/A_{vdd}$, where $A_v = v_{out}/v_{in}$, $A_{vdd} = v_{out}/v_{dd}$).

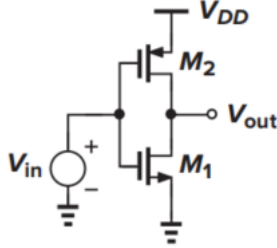


Figure 4(a)

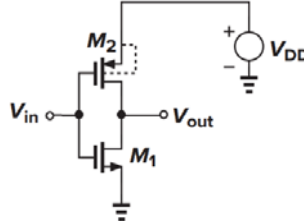


Figure 4(b)

5 Source degeneration

- Consider the circuit of Figure 5(a) with $(W/L)_1 = 50/0.5$, $R_D = 2k\Omega$, $R_S = 200\Omega$,
- Calculate the equivalent transconductance (G_m) with the assistance of the small-signal model in Figure 5(b).
 - Calculate the small-signal voltage gain if $I_D = 0.5mA$, assuming $\lambda = \gamma = 0$.
 - Assuming $\lambda = \gamma = 0$, calculate the input voltage that places M_1 at the edge of the triode region. What is the gain under this condition?

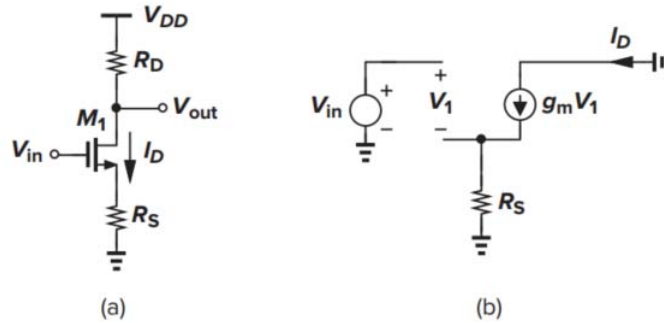


Figure 5

Source Follower

6. For the common-drain amplifier (source follower) in the Figure 6, assume $(W/L)=10$ and $\lambda=0$. Find the dc output voltage V_o and the small-signal gain v_o/v_i under the following conditions: $\mu_n C_{ox} = 90\mu A/V^2$, $\gamma = 0.19\sqrt{V}$, $2\phi_F = 0.6V$
- Ignoring the body effect and with $R \rightarrow \infty$.
 - Including the body effect and with $R \rightarrow \infty$.
 - Including the body effect and with $R = 100k\Omega$.
 - Including the body effect and with $R = 10k\Omega$.
- (Note: the V_{TH} in the question (b) can be directly used in (c) and (d) without calculating it again).

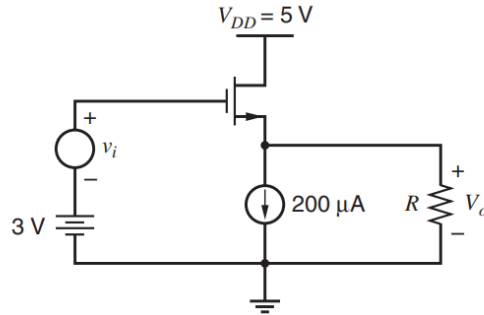


Figure 6

Common Gate Stage

7. Determine the input resistance of the CG amplifier of the Figure 8, if the transistor operates in the active region with $I_D = 100\mu A$. Let $R_D = 10k\Omega$, $\mu_n C_{ox} = 200\mu A/V^2$, $\lambda = 0.01V^{-1}$, $W = 100\mu m$, and $L = 1\mu m$. Ignore the body effect. Repeat with $R_D = 1M\Omega$. If the $100\mu A$ current flows through R_D in this case, a power-supply voltage of at least 100V would be required. To overcome this problem, assume that an ideal $100\mu A$ current source is placed in parallel with R_D here.

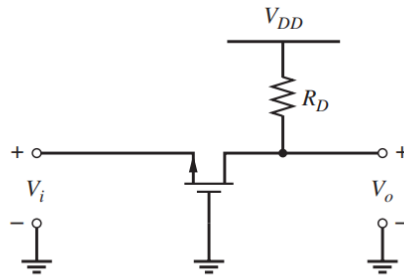


Figure 7

Cascode Stage

8. In the cascode stage of the Figure 8, assume that $(W/L)_1 = 50/0.5$, $(W/L)_2 = 10/0.5$, $I_{D1} = I_{D2} = 0.5mA$ and $R_D = 1k\Omega$.

- Choose V_b such that M_1 is 50mV away from the triode region.
- Calculate the small-signal voltage gain.
- Using the value of V_b found in part (a), calculate the maximum output voltage swing. Which device enters the triode region first as V_{out} falls?
- Calculate the swing at node X for the maximum output swing obtained above.

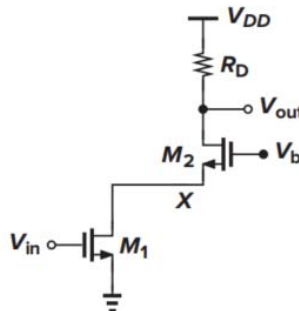


Figure 8

9. Figure 9 shows a folded-cascode CMOS amplifier. The DC current in the current source transistor M2 is $2I$, and the current in the cascode current source (M4 and M5) is I . For simplicity, assume that all transistors have equal parameters g_m and r_o , and $1/g_m \ll r_o$, ignore the body effect of all transistors. Answer the following questions using only g_m and r_o :

- Give approximate expressions for all the small-signal resistances indicated in Figure 10, i.e. $R_{o1} \sim R_{o5}$ and R_{in3} .
- Find the amplifier output resistance R_o .
- Verify that the short-circuit transconductance G_m is approximately equal to g_{m1} . Note that the short-circuit transconductance is determined by short-circuiting v_o to AC ground and finding the current that flows through the short circuit, $G_m v_i$.
- Find the overall voltage gain v_o/v_i and evaluate its value for the case $g_m = 2 \text{ mA/V}$ and $r_o = 15 \text{ K}\Omega$.

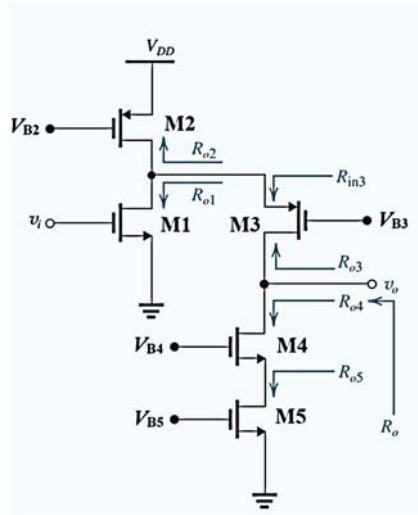


Figure 9

Comprehensive Application Problem

10. For the circuit of Fig. 10, $\mu_n C_{OX}(W/L)_1 = \mu_p C_{OX}(W/L)_2 = 1 \text{ mA/V}^2$, $V_{TH1} = |V_{TH2}| = 0.5 \text{ V}$. The resistance $R = 1 \text{ M}\Omega$, and $V_{DD} = 2 \text{ V}$. Ignore the body effect and answer the following questions:

- For G and D open, what are the drain currents I_{D1} and I_{D2} ?
- For $r_{o1} = r_{o2} = \infty$, what is the voltage gain of the amplifier from G to D? (Hint: using small-signal models for calculation)
- For finite r_{o1} and r_{o2} ($\lambda_1 = \lambda_2 = 0.05 \text{ V}^{-1}$), what is the voltage gain from G to D and the small-signal input resistance at G?
- If G is driven from a signal source v_{sig} (having an internal resistance of $20 \text{ k}\Omega$), find the voltage gain v_d/v_{sig} .
- For what range of output signals do M1 and M2 remain in the saturation region? (HINT: the DC levels at D and G are equal for all the questions)

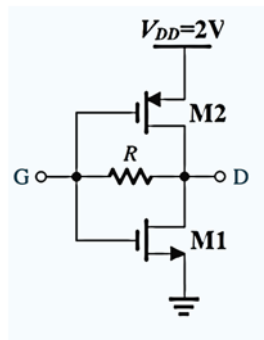


Figure 10

Appendix

The process parameters may be used in all the questions are shown in the table below. To simplify the calculation, you can use the calculation results of $\mu_n C_{ox}$ and $\mu_p C_{ox}$ below.

If some similar parameters are already given in the question, you should use those parameters given in the question directly.

If the value of V_{DD} is not given in the question, use $V_{DD}=3V$ as a default value.

Table 2.1 Level 1 SPICE models for NMOS and PMOS devices.

NMOS Model			
LEVEL = 1	VTO = 0.7	GAMMA = 0.45	PHI = 0.9
NSUB = 9e+14	LD = 0.08e-6	UO = 350	LAMBDA = 0.1
TOX = 9e-9	PB = 0.9	CJ = 0.56e-3	CJSW = 0.35e-11
MJ = 0.45	MJSW = 0.2	CGDO = 0.4e-9	JS = 1.0e-8
PMOS Model			
LEVEL = 1	VTO = -0.8	GAMMA = 0.4	PHI = 0.8
NSUB = 5e+14	LD = 0.09e-6	UO = 100	LAMBDA = 0.2
TOX = 9e-9	PB = 0.9	CJ = 0.94e-3	CJSW = 0.32e-11
MJ = 0.5	MJSW = 0.3	CGDO = 0.3e-9	JS = 0.5e-8

$$\begin{aligned}\mu_n C_{ox} &= \mu_n \times \frac{\epsilon_o \epsilon_{ox}}{t_{ox}} \\ &= 350 \times \frac{8.85 \times 10^{-12} \times 3.9}{9 \times 10^{-9}} \\ &\approx 1.34225 \times 10^{-4} \text{ A/V}^2\end{aligned}$$

$$\begin{aligned}\mu_p C_{ox} &= \mu_p \times \frac{\epsilon_o \epsilon_{ox}}{t_{ox}} \\ &= 100 \times \frac{8.85 \times 10^{-12} \times 3.9}{9 \times 10^{-9}} \\ &\approx 0.3835 \times 10^{-4} \text{ A/V}^2\end{aligned}$$